

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Built **Python** backends for **AI-driven features (Agents SDK, MCP)** at **millions of daily transactions, 99%+ uptime**.
- Deployed **AWS services (Lambda, API Gateway, SageMaker)** interfacing with **ML platforms** at **sub-200ms latency**.
- Built search and retrieval infrastructure on **Elasticsearch** with **knowledge graph** enrichment behind typed **REST APIs**.
- Built per-node **Redis checkpointing** so interrupted distributed workflows **resume from the last committed step**.
- Replaced polling with **SSE** for **real-time AI analytics** on live ML metrics, cutting redundant API calls **90%**.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Cut **p99 inference latency 420ms to 140ms** across **50K+ daily requests** via async **vLLM** on **GCP Cloud Run**.
- Designed **BigQuery** schemas for **AI data workloads**, cutting compute cost via **partition pruning** and **clustering**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Architected **25+ Java/Python microservices** for **1M+ users** at **99.9% uptime** over 4 years on **AWS** (EC2, RDS, SQS).
- Cut nightly financial aggregation blocking downstream settlement from **8 minutes to 90 seconds** by redesigning SQL with **indexed views, batch partitioning, and lock contention** elimination.
- Designed **Python/C++** model-serving infrastructure with **async request handling, batching**, health checks, and retry policies; raised ML service availability to **99.9%** and cut **p99 latency 32%**.
- Drove **backend performance optimization** of request scheduling and concurrency: **2.4x** throughput, **27%** less GPU memory.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- **Raft-consensus** KV store with **linearizable reads** and **log compaction**; companion **VPC control plane** simulator.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- **LLM eval platform** with **drift detection** and latency tracking; **FastAPI** backend, **React/TypeScript, PostgreSQL**.

LangFetch

tinyurl.com/LangFetch

- **LangGraph** agent decomposing questions into dependency-ordered **SQL subqueries**; **MCP-compatible** server.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, Java, C++, Go, SQL, TypeScript, JavaScript, HTML/CSS

Backend & Data: FastAPI, Flask, REST APIs, Microservices, PostgreSQL, Redis, Elasticsearch, DynamoDB, NoSQL, BigQuery, Kafka, SQS

Cloud & Infra: AWS (EC2, RDS, Lambda, S3, SageMaker, ECS, EKS, SQS), GCP (Cloud Run), Docker, Kubernetes, Terraform, GitHub Actions, CI/CD

AI/ML & Fundamentals: vLLM, PyTorch, RAG, LangGraph, MCP; Distributed Systems, Database Systems, Data Structures & Algorithms, Operating Systems

CERTIFICATIONS

AWS Certified Developer Associate | GCP Professional MLOps Engineer